

Superseding the selection principle for quantitative Saint–Venant via the optimal radial graph: cut, do not fill

Abstract

To remove the Brasco–De Philippis–Velichkov selection step it suffices, as observed by the author, to produce from a near-extremal Ω a *radial graph* G with $|\Omega \triangle G| = O(\delta_T(\Omega))$; the small-amplitude radial-graph theorem (G) (a local, regularity-free inequality) then closes the estimate. The natural candidate — the radial hull (star-shaped envelope) — fails, because it can only *fill* pockets, and a thin overhang has pocket volume $\gg \delta_T$ (no-overhang obstruction). The resolution is to allow the graph to *cut* as well as fill: the *optimal radial graph*, which removes thin caps and fills thin gaps, has defect

$$\text{gd}(\Omega) := \min_{G \text{ radial graph}} |\Omega \triangle G| \lesssim \eta(\delta_{\text{iso}} + \eta),$$

where η is the annular half-width and δ_{iso} the isoperimetric deficit (Theorem 4). *No no-overhang hypothesis appears*; overhangs are cut away at a cost charged to δ_{iso} . The radial-graph theorem needs only amplitude $\leq$ a fixed constant, which is automatic; so the realization imposes *no width condition*. Consequently:

- the entire sharp theorem reduces, with selection deleted, to producing a graph whose volume, deficit, and closeness to Ω are controlled — and volume and closeness are free, so the *only* constraint is that the surgical repair shift the deficit by $O(\delta_T)$, which holds once $\eta \lesssim \delta_T^{1/4}$ (not $\sqrt{\delta_T}$; an earlier draft overstated this) (Theorem 7);
- unconditionally one gets the selection-free $\mathcal{A}_F(\Omega)^2 \lesssim \delta_T^{2/3}$ — with no inclusion at all, by applying quantitative isoperimetry to the inset, whose perimeter (unlike $\partial\Omega$'s) is controlled by δ_T (Proposition 13).

We pursue two routes to the graph. *Route 2* repairs the inset by cut-or-fill (the above). *Route 1* asks whether the inset is *already* a graph, with no repair; we prove the *folding identity* $\int_0^\infty F(t) dt = \int_\Omega (\partial_r u)_+ dx$ (with F the folded radial perimeter of the level set $\{u = t\}$), which shows the inset is a graph iff $\partial_r u \leq 0$ in the boundary layer, settles the interior unconditionally, and pins the obstruction to a single sign condition on $\partial_r u$ at the boundary — the same quantity $M = \int (\partial_r u)_+$ that Route 2 pays for. We give complete proofs of both reductions and the folding identity, isolate the two remaining inputs (Route 2: a $\delta_T^{1/4}$ Hausdorff annulus; Route 1: radial monotonicity), and record the overhang-free Steiner-flow alternative.

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1 Architecture and the target

Work in $\mathbb{R}^2$. For bounded Ω of finite measure let u_Ω be its torsion function ($-\Delta u_\Omega = 1$, $u_\Omega \in H_0^1(\Omega)$), $T(\Omega) = \int_\Omega u_\Omega$, B the disk of area $|\Omega|$, and

$$\delta_T(\Omega) := T(B) - T(\Omega) \geq 0, \quad \mathcal{A}_F(\Omega) := \inf_x \frac{|\Omega \triangle B(x)|}{|\Omega|}, \quad |B(x)| = |\Omega|.$$

Normalise $|\Omega| = \pi$, $B = B_1$, $T(B_1) = \pi/8$. The sharp goal is

$$\mathcal{A}_F(\Omega)^2 \leq C \delta_T(\Omega) \quad (\delta_T(\Omega) \leq \delta_\star). \quad (1)$$

The reduction target (the user's observation). The selection step exists only to manufacture a $C^{1,\gamma}$ nearly-spherical representative on which two regularity-dependent moves run: an annular inclusion and a radial-graph realization. But for (1) one needs much less: *any* radial graph G close to Ω in measure suffices, because the radial-graph theorem is intrinsic.

It is enough to produce a radial graph G with $|\Omega \triangle G| = O(\delta_T(\Omega))$; then a local analysis (the radial-graph theorem) finishes.

We make this precise (§2), then show the target is attainable by the *optimal* radial graph, with the no-overhang obstruction removed by allowing cuts (§3–4).

Module (G), imported. *Radial-graph theorem.* If $G = \{r < 1 + \varphi(\omega)\}$ with $\varphi \in C^0(S^1)$, $\|\varphi\|_{L^\infty} \leq \eta_G$, $|G| = \pi$, barycentre 0, then $\mathcal{A}_F(G)^2 \leq C_G \delta_T(G)$ (planar constants $\eta_G = 1/13$, $C_G < 48$; manuscript §Step 4). This module uses no property of the original Ω and no selection.

Deficits and scales. After the level-set reduction (manuscript §Step 2), at the stopped level $\hat{t} \sim \sqrt{\delta_T}$ the controlling deficits are the Serrin defect and the isoperimetric deficit of the inset, with

$$\varepsilon := \left(\int_{\partial\Omega} (|\nabla u_\Omega| - c_\Omega)^2 \right)^{1/2} \sim \delta_T^{1/4}, \quad \delta_{\text{iso}} := \frac{P(\Omega)}{2\sqrt{\pi|\Omega|}} - 1 \sim \delta_T^{1/2}. \quad (2)$$

2 The reduction: a graph within δ_T suffices

Lemma 1 (Boundary-layer torsion stability). *Let $\Omega, G \subset B_{1+\eta}(z)$ both contain $B_{1-\eta}(z)$, with $|\Omega \triangle G| = m$ (so $\Omega \triangle G$ lies in the collar $\{1 - \eta < |x - z| < 1 + \eta\}$). Then*

$$|T(\Omega) - T(G)| \leq 3\eta m. \quad (3)$$

Proof. Put $E := \Omega \cap G \subset F := \Omega \cup G$, $|F \setminus E| = m$, so $|T(\Omega) - T(G)| \leq T(F) - T(E)$. Writing $T(D) = \iint_{D \times D} G_D(x, y) dx dy$ with $G_D \geq 0$ the Green function and using the monotonicity $G_E \leq G_F$,

$$T(F) - T(E) \leq \iint_{F \times F} G_F - \iint_{E \times E} G_F = 2 \iint_{E \setminus F \setminus E} G_F + \iint_{(F \setminus E)^2} G_F.$$

For $y \in F \setminus E$ (in the collar), the comparison $u_F \leq u_{B_{1+\eta}}$ gives $u_F(y) = \int_F G_F(\cdot, y) \leq ((1 + \eta)^2 - (1 - \eta)^2)/4 = \eta$. Hence $\int_E \int_{F \setminus E} G_F \leq \int_{F \setminus E} u_F \leq \eta m$ and $\int \int_{(F \setminus E)^2} G_F \leq \int_{F \setminus E} u_F \leq \eta m$, giving $T(F) - T(E) \leq 3\eta m$. $\square$

Theorem 2 (Reduction — exactly the three controlled quantities). *Let $\Omega \subset \mathbb{R}^2$, $|\Omega| = \pi$, $\delta_T(\Omega) \leq \delta_*$. Suppose there is a radial graph $G = \{z + r\omega : r < \rho(\omega)\}$, $\rho \in C^0(S^1)$, with*

- (i) (graph, amplitude a fixed constant) $\|\rho - 1\|_{L^\infty} \leq \eta_G$ and barycentre at z , where η_G is the absolute constant of (G) ($= 1/13$);
- (ii) (volume controlled) $||G| - \pi| \leq K\delta_T(\Omega)$;
- (iii) (deficit controlled) $\delta_T(G) \leq K\delta_T(\Omega)$;
- (iv) (closeness) $|\Omega \triangle G| \leq K\sqrt{\delta_T(\Omega)}$.

Then $\mathcal{A}_F(\Omega)^2 \leq C(K, C_G)\delta_T(\Omega)$, with no selection and no regularity of $\partial\Omega$.

Remark 3 (On hypothesis (i): the amplitude is free). The graph theorem (G) requires only $\|\rho - 1\|_{L^\infty} \leq \eta_G$ with η_G a fixed absolute constant; it does *not* require an amplitude that shrinks with δ_T . Since $\delta_{\text{iso}} \rightarrow 0$ forces the inset into an annulus of width $o(1)$, (i) holds automatically for $\delta_T \leq \delta_*$. **There is no “ $\sqrt{\delta_T}$ -width annulus” requirement**; an earlier draft wrongly imposed one. The substance is (iii)+(iv): a graph whose deficit and symmetric difference are controlled.

Proof. Rescale $\tilde{G} := sG$, $s = (\pi/|G|)^{1/2}$, so $|\tilde{G}| = \pi$; by (ii) $s^4 = 1 + O(\delta_T)$, so $\delta_T(\tilde{G}) \leq \delta_T(G) + C\delta_T \leq (K + C)\delta_T$ using (iii). Recentre the barycentre (translation $O(\eta_G)$). By (i), $\tilde{G}$ satisfies the hypotheses of (G), giving $\mathcal{A}_F(\tilde{G})^2 \leq C_G\delta_T(\tilde{G}) \leq C_G(K + C)\delta_T$. Finally $\mathcal{A}_F(\tilde{G}) = \mathcal{A}_F(G)$ (dilation invariance) and, by (iv) and $\delta_T \leq 1$,

$$\mathcal{A}_F(\Omega) \leq \mathcal{A}_F(G) + \frac{|\Omega \triangle G|}{\pi} \leq \sqrt{C_G(K + C)\delta_T} + \frac{K}{\pi}\sqrt{\delta_T} \leq C(K, C_G)\delta_T^{1/2}. \quad \square$$

This is precisely your statement: *produce a radial graph with controlled volume and deficit (and, automatically, closeness to Ω), and we are done.* The amplitude in (i) is a free constant. The rest of the note constructs such a G and pins down what (iii) costs.

3 Why the radial hull fails, and why cutting succeeds

Assume the inclusion $B_{1-\eta}(0) \subset \Omega \subset B_{1+\eta}(0)$ (input (A), §5); centre 0. For a.e. θ the slice $\Omega_\theta = \{r : re^{i\theta} \in \Omega^{(1)}\}$ is, by Vol’pert slicing, a finite union of intervals

$$\Omega_\theta = (0, r_1) \cup (r_2, r_3) \cup \cdots \cup (r_{2m}, r_{2m+1}), \quad r_- \leq r_1 < \cdots < r_{2m+1} \leq r_+,$$

($r_\pm = 1 \pm \eta$, $m = m(\theta)$). The *gaps* (r_{2i-1}, r_{2i}) and *caps* (r_{2i}, r_{2i+1}) are the obstruction to Ω being a graph.

The hull only fills. The radial hull $G_{\text{hull}} = \{r < R(\theta)\}$, $R = r_{2m+1}$, fills every gap; its defect is $|G_{\text{hull}} \setminus \Omega| = \sum_\theta \sum_i$ (gap masses). A thin overhang — a wide cap on a thin neck (lollipop) — has a deep gap beneath it: gap mass $\sim \eta$ while $\delta_T \rightarrow 0$ (`density_to_graph_realization.tex`, Prop. “floating sliver”). So $|G_{\text{hull}} \triangle \Omega| \gg \delta_T$. The hull therefore needs a no-overhang hypothesis, which neither finite perimeter nor density supplies; this was the barrier.

Cutting dissolves the barrier. Replace “fill” by “cut-or-fill, whichever is cheaper, in each slice.” Define the *optimal radial graph defect*

$$\text{gd}(\Omega) := \inf_{\rho: S^1 \rightarrow [r_-, r_+]} \int_{S^1} \int_0^\infty |\mathbf{1}_\Omega(re^{i\theta}) - \mathbf{1}_{\{r < \rho(\theta)\}}| r dr d\theta = \int_{S^1} \left(\min_\rho |\Omega_\theta \triangle (0, \rho)|_r \right) d\theta, \quad (4)$$

($|\cdot|_r$ the 1-D mass with weight $r dr$). Per slice the optimal $\rho(\theta)$ either *cuts* the caps (take $\rho = r_1$) or *fills* the gaps (take $\rho = r_{2m+1}$) or stops in between — it picks the minority. For the lollipop one simply *cuts* the wide cap: the cap has mass $\sim \delta_T$ (it is the displaced, asymmetric material), so the defect is $\sim \delta_T$, not $\sim \eta$. The deep gap is never filled. The no-overhang obstruction does not arise because we are never forced to fill a deep gap.

4 The graph defect is controlled by $\eta \delta_{\text{iso}}$

We now bound gd by the geometry alone. The mechanism: each fold costs perimeter proportional to its angular width, so the total fold-arc is $\lesssim \delta_{\text{iso}}$; cutting the caps then costs $\leq \eta$ per unit fold-arc.

Theorem 4 (Graph-defect bound). *Let $\Omega \subset \mathbb{R}^2$ be of finite perimeter, $|\Omega| = \pi$, with $B_{1-\eta}(0) \subset \Omega \subset B_{1+\eta}(0)$, $\eta \leq \frac{1}{4}$, and isoperimetric deficit δ_{iso} . Then*

$$\text{gd}(\Omega) \leq C \eta (\delta_{\text{iso}} + \eta), \quad C \leq 40. \quad (5)$$

Moreover the optimal profile ρ may be taken in $BV(S^1)$ with $\rho \in [r_-, r_+]$, and a continuous $\tilde{\rho} \geq \rho - C\eta \delta_{\text{iso}}$ with $|\{\tilde{\rho} \neq \rho\}\text{-region}| \leq C\eta(\delta_{\text{iso}} + \eta)$ exists, so the realizing graph in Theorem 2 can be taken continuous at the same cost.

Proof. Decompose the reduced boundary into radial-normal arcs ($\nu = \pm e_r$, “caps/floors”) and angular-normal walls ($\nu = \pm e_\theta$, “radial walls”), $P(\Omega) = P_r(\Omega) + P_\theta(\Omega)$ (an equality up to the $\mathcal{H}^1$ -null tangential set; in general $P \geq \max(P_r, P_\theta)$ and $P \geq \frac{1}{\sqrt{2}}(P_r + P_\theta)$, which is all we use).

Step 1: fold count from P_r . By the radial slicing formula [1, Thm. 3.103],

$$P_r(\Omega) = \int_{S^1} \sum_{k=1}^{2m(\theta)+1} r_k(\theta) d\theta \geq \int_{S^1} (2m(\theta) + 1) r_- d\theta = r_- \left(2 \int_{S^1} m d\theta + 2\pi \right).$$

On the other hand $P_r(\Omega) \leq P(\Omega) = 2\pi(1 + \delta_{\text{iso}})$. Hence

$$\int_{S^1} m d\theta \leq \frac{\pi(1 + \delta_{\text{iso}})}{r_-} - \pi = \pi \frac{1 + \delta_{\text{iso}} - r_-}{r_-} \leq \frac{\pi(\delta_{\text{iso}} + \eta)}{1 - \eta} \leq 2\pi(\delta_{\text{iso}} + \eta), \quad (6)$$

using $r_- = 1 - \eta$ and $\eta \leq \frac{1}{4}$. This bounds the total fold-arc: $\int m d\theta = \mathcal{H}^1\{(\theta, i) : i\text{-th fold present at } \theta\}$.

Step 2: cut the caps. Take the core profile $\rho_0 := r_1(\theta)$ (cut every cap). Then $\Omega \triangle \{r < \rho_0\} = \Omega \setminus \{r < \rho_0\} = \text{caps}$, and the per-slice cap mass is

$$\sum_{i=1}^m \frac{r_{2i+1}^2 - r_{2i}^2}{2} \leq r_+ \sum_{i=1}^m (r_{2i+1} - r_{2i}) \leq r_+ (r_+ - r_-) \min(1, m) \leq 2\eta r_+ \mathbf{1}_{\{m \geq 1\}},$$

since the caps lie in $[r_-, r_+]$ of total radial width $\leq r_+ - r_- = 2\eta$. Therefore, with $\mathbf{1}_{\{m \geq 1\}} \leq m$,

$$\text{gd}(\Omega) \leq \int_{S^1} 2\eta r_+ \mathbf{1}_{\{m \geq 1\}} d\theta \leq 2\eta r_+ \int_{S^1} m d\theta \leq 2\eta \cdot \frac{5}{4} \cdot 2\pi(\delta_{\text{iso}} + \eta) \leq 40\eta(\delta_{\text{iso}} + \eta). \quad (7)$$

This is (5).

Step 3: continuity at the same cost. $\rho_0 = r_1 \in BV(S^1)$ with $\rho_0 \in [r_-, r_+]$; its jump set has total jump $\leq 2\eta \cdot \#\{\text{jumps}\}$ and total variation $\text{TV}(\rho_0) \leq P_\theta(\Omega) \leq P(\Omega)$, but more usefully the number of jumps exceeding height h is $\leq \text{TV}/h$. Mollify ρ_0 at angular scale σ : the smoothed $\tilde{\rho}_0$ differs from ρ_0 on a set of angular measure $\leq \text{TV}(\rho_0)/(\cdot)$ and adds area $\leq \sigma \text{TV}(\rho_0) r_+$; choosing σ so that $\sigma \text{TV}(\rho_0) \leq \eta \delta_{\text{iso}}$ keeps the extra area $\leq \eta \delta_{\text{iso}} r_+ \leq C\eta(\delta_{\text{iso}} + \eta)$. The mollified profile is Lipschitz, $\tilde{\rho}_0 \in [r_-, r_+]$, and $|\Omega \Delta \{r < \tilde{\rho}_0\}| \leq \text{gd}(\Omega) + C\eta(\delta_{\text{iso}} + \eta) \leq C'\eta(\delta_{\text{iso}} + \eta)$. $\square$

Remark 5 (On the η^2 term). The bound (5) carries an η^2 from the slack $\int m \leq 2\pi(\delta_{\text{iso}} + \eta)$ in (6) (the inclusion contributes $r_- = 1 - \eta < 1$). For a single fold of arc ℓ and depth $d \leq \eta$ one has directly $\text{gd} \sim \ell d$ and $\delta_{\text{iso}} \sim \ell$, i.e. $\text{gd} \sim \eta \delta_{\text{iso}}$ with *no* η^2 ; the η^2 is therefore not attained when folds are scarce ($\delta_{\text{iso}} \ll \eta$, the relevant regime) and is an artifact. In the sharp regime $\eta \lesssim \delta_{\text{iso}}$ it is harmless ($\eta^2 \leq \eta \delta_{\text{iso}}$). We keep it because removing it cleanly requires the near-ball normalisation $r_\pm \rightarrow 1$, which the inclusion does not quite provide.

4.1 Which of the four conditions cost what

Take G to be the *volume-preserving* optimal radial graph: choose ρ with $\frac{1}{2} \int \rho^2 = \pi$ minimising the defect; then $|G| = \pi$ exactly, $\rho \in [r_-, r_+]$, and (by cutting the caps as in Theorem 4, using the clean $\text{gd} \lesssim \eta \delta_{\text{iso}}$ form — the η^2 term is an artifact, Remark 5) $|\Omega \Delta G| \leq C\eta \delta_{\text{iso}}$. We check the four hypotheses of Theorem 2:

- (i) amplitude $\|\rho - 1\|_{L^\infty} \leq \eta \leq \eta_G$: **free** (holds once $\eta \leq 1/13$, automatic);
- (ii) volume $|G| = \pi$: **free** (volume-preserving choice);
- (iv) closeness $|\Omega \Delta G| \leq C\eta \delta_{\text{iso}} \leq C\delta_{\text{iso}} \leq C\sqrt{\delta_T}$: **free** (since $\eta \leq 1$);
- (iii) deficit: by Lemma 1, $\delta_T(G) \leq \delta_T(\Omega) + 3\eta |\Omega \Delta G| \leq \delta_T(\Omega) + 3C\eta^2 \delta_{\text{iso}}$. This is the *only* condition that constrains η .

So three of the four are free; the surgery used to manufacture the graph shifts the deficit by $\lesssim \eta^2 \delta_{\text{iso}}$, and that is the entire price.

Corollary 6 (The surgery's deficit cost). *With $\delta_{\text{iso}} \sim \sqrt{\delta_T}$, the deficit shift is $3C\eta^2 \delta_{\text{iso}} \leq \delta_T(\Omega)$ iff*

$$\eta \lesssim \sqrt{\delta_{\text{iso}}} \sim \delta_T^{1/4}. \quad (8)$$

Proof. $\eta^2 \delta_{\text{iso}} \lesssim \delta_T \sim \delta_{\text{iso}}^2 \iff \eta^2 \lesssim \delta_{\text{iso}}$. $\square$

5 Route 2 (surgery): the main reduction, selection-free

Theorem 7 (Selection deleted). *Let $\Omega \subset \mathbb{R}^2$, $|\Omega| = \pi$, $\delta_T(\Omega) \leq \delta_*$, with $\delta_{\text{iso}} \sim \sqrt{\delta_T}$. Suppose input (A): a centre z and an inclusion $B_{1-\eta}(z) \subset \Omega \subset B_{1+\eta}(z)$ with*

$$\eta \lesssim \sqrt{\delta_{\text{iso}}} \quad (\sim \delta_T^{1/4}). \quad (9)$$

Then $\mathcal{A}_F(\Omega)^2 \leq C \delta_T(\Omega)$, with no selection principle, no $C^{1,\gamma}$ regularity, and no no-overhang hypothesis.

Proof. The volume-preserving optimal radial graph G of §4.1 satisfies (i),(ii),(iv) of Theorem 2 unconditionally; by Corollary 6 and (9) it satisfies (iii), $\delta_T(G) \leq 2\delta_T(\Omega)$. Taking ρ continuous (Theorem 4, Step 3, at cost $\leq C\eta \delta_{\text{iso}}$ to the defect, absorbed) and applying Theorem 2 gives the claim. $\square$

Remark 8 (The width that is *not* needed, and the one that is). The graph realization itself imposes *no* width condition: a graph of amplitude $\leq \eta_G$ (a fixed constant) is what (G) wants, and that is automatic. The condition (9), $\eta \lesssim \delta_T^{1/4}$, comes *solely* from keeping the deficit of the surgically repaired graph within $O(\delta_T)$ — hypothesis (iii). It is much weaker than the $\sqrt{\delta_T}$ -width annulus an earlier draft demanded. If the inset were already a graph (no surgery, $gd = 0$), even this would be vacuous and the result would be unconditional.

Remark 9 (What this accomplishes). The selection-based proof used $C^{1,\gamma}$ for two things: the BNST annular inclusion *and* the radial-graph realization (pocket lemma). Theorem 7 eliminates the second entirely — the optimal radial graph realizes Ω within δ_T *from finite perimeter alone* — and shows the first is the *only* remaining input. Equivalently: **the no-overhang obstruction, which blocked the density-to-hull route, is an artifact of insisting on the hull; cut-or-fill removes it.** What is left is a single, classical-looking estimate: an annulus of Hausdorff width $\lesssim \delta_T^{1/4}$.

6 Route 1 (no surgery): the folding identity and radial monotonicity

Route 2 *repairs* the inset into a graph and pays a deficit cost. Route 1 asks whether the inset *is already* a graph, so that $\delta_T(G) \leq 4\delta_T$ holds with no repair and the result is unconditional. We give the exact criterion and a clean identity that measures the failure.

Throughout $u = u_\Omega$, $E_t = \{u > t\}$, and 0 is the chosen radial centre. Where $\nabla u \neq 0$ the outward unit normal of E_t is $\nu_t = -\nabla u / |\nabla u|$, so

$$\nu_t \cdot e_r = -\frac{\partial_r u}{|\nabla u|}, \quad \partial_r u := \nabla u \cdot e_r.$$

E_t is a *radial graph* about 0 (star-shaped, no overhang) iff $\nu_t \cdot e_r \geq 0$ $\mathcal{H}^1$ -a.e. on $\partial^* E_t$, i.e. iff $\partial_r u \leq 0$ there. Define the *folded radial perimeter*

$$F(t) := \int_{\partial^* E_t} (\nu_t \cdot e_r)_- d\mathcal{H}^1 = \int_{\partial^* E_t} \frac{(\partial_r u)_+}{|\nabla u|} d\mathcal{H}^1 \geq 0, \quad F(t) = 0 \iff E_t \text{ is a radial graph.}$$

Lemma 10 (Folding identity). *For the torsion function of any bounded set Ω of finite perimeter,*

$$\int_0^\infty F(t) dt = \int_\Omega (\partial_r u)_+ dx =: M(\Omega). \quad (10)$$

Consequently:

- (i) *if $\partial_r u \leq 0$ a.e. in the annular layer $\{0 < u < t_0\}$, then every level set there is a radial graph (Route 1 applies, with no surgery);*
- (ii) *$M(\Omega)$ is the total non-graphness; by averaging, some level in any sub-layer of width ℓ has $F(t) \leq M/\ell$.*

Proof. $(\nu_t \cdot e_r)_- = (\partial_r u)_+ / |\nabla u|$ pointwise on $\partial^* E_t$. The coarea formula $\int_0^\infty \int_{\partial^* E_t} \frac{g}{|\nabla u|} d\mathcal{H}^1 dt = \int_\Omega g dx$ with $g = (\partial_r u)_+$ gives (10). (i),(ii) are immediate. $\square$

Lemma 11 (The interior is graph-like for free). *There are $t_0 > 0$ and $\delta_0 > 0$ (absolute) such that if $\delta_T(\Omega) \leq \delta_0$ and Ω is suitably centred, then $\partial_r u < 0$ on $\{u \geq t_0\}$; hence every inner level set $\{u = t\}$, $t \geq t_0$, is a radial graph.*

Proof sketch. $|\nabla u| \leq C$ (global gradient bound), so $u \leq C \text{dist}(\cdot, \partial\Omega)$ and $\{u \geq t_0\} \subset \{\text{dist} \geq t_0/C\}$. On this interior region $-\Delta u = 1$ gives, by interior Schauder estimates, C^2 bounds depending only on t_0 and $\text{diam } \Omega$. If $\delta_T(\Omega_k) \rightarrow 0$ then $T(\Omega_k) \rightarrow T(B)$, and by the rigidity of Saint-Venant together with elliptic compactness $u_k \rightarrow u_{B_1}$ in C_{loc}^1 of the interior (after recentring); since $\partial_r u_{B_1} = -r/2 < 0$ there, $\partial_r u_k < 0$ on $\{u_k \geq t_0\}$ for k large, i.e. $\delta_T \leq \delta_0$. $\square$

The crux of Route 1. Lemma 11 settles the *interior*: the obstruction to Route 1 is confined to the boundary layer $\{0 < u < t_0\}$, and is precisely a *sign condition*,

$$\partial_r u \leq 0 \quad \text{on} \quad \{0 < u < t_0\} \quad (\text{radial monotonicity in the layer}). \quad (11)$$

This is exactly what fails at an overhang of $\partial\Omega$: beneath an outward cap, u *increases* outward ($\partial_r u > 0$) toward the cap. A $C^{1,\gamma}$ boundary (what selection delivers) controls ∇u up to $\partial\Omega$ and gives (11) for δ_T small; with only finite perimeter it can fail, and then $M(\Omega) > 0$.

Remark 12 (Route 1 $\Leftrightarrow$ Route 2, quantitatively). The folding identity unifies the two routes. If (11) holds, $M = 0$, every layer level is a graph, and Route 1 finishes with no deficit cost. If not, $M > 0$ measures the total fold; the level-set deficits bound it,

$$M(\Omega) = \int_0^\infty F(t) dt \leq \frac{1}{2} \int_0^\infty (P_t - I_t) dt, \quad I_t := \int_{E_t} \frac{1}{r} dx, \quad (12)$$

since $P_t = \mathcal{H}^1(\partial^* E_t) \geq \int |\nu_t \cdot e_r| = I_t + 2F(t)$ (the identity $\int_{\partial^* E_t} \nu_t \cdot e_r = \int_{E_t} \operatorname{div} e_r = I_t$). Each fold is then *cut* at its level (Route 2). Thus the same quantity $M = \int (\partial_r u)_+$ governs both: Route 1 is the case $M = 0$, Route 2 pays $\propto M$. The honest status: $M = 0$ (pure Route 1) is the radial-monotonicity (11), of the same strength as the boundary regularity selection provides; absent it, Route 2 with $\eta \lesssim \delta_T^{1/4}$ is the fallback.

7 The remaining input

The two routes leave two clearly different open inputs:

- **Route 1** needs the sign condition (11), $\partial_r u \leq 0$ in the boundary layer (radial monotonicity), i.e. $M(\Omega) = 0$ — a regularity-flavoured statement about ∇u up to $\partial\Omega$.
- **Route 2** needs the inclusion (9), a *Hausdorff* annulus width $\eta \lesssim \delta_T^{1/4}$.

We discuss Route 2's input, the more classical of the two. The hypothesis (9) is an overdetermined-stability statement: *small Serrin deficit (and small δ_{iso}) $\Rightarrow \Omega$ trapped in an annulus of Hausdorff width $\lesssim \delta_T^{1/4}$* . This is the *Hausdorff* (pointwise) width, the quantity that resists overhangs — genuinely stronger than the L^1 closeness $|\Omega \triangle B| \lesssim \sqrt{\delta_{\text{iso}}}$ that quantitative isoperimetry gives directly. Two sub-routes:

(a) With selection, this is BNST. For $C^{1,\gamma}$ insets the Brandolini–Nitsch–Salani–Trombetti theorem gives an annulus of width $O(\delta_T^\mu)$, sharpened after the $C^{1,\gamma}$ inset; this is the manuscript's Step 2. It uses the regularity that selection provides.

(b) Without selection: Magnanini–Poggesi under density. The flux mechanism of `almost_serrin_discussion.tex` gives two-sided density above scale $r_* \sim \varepsilon^2 + \delta_{\text{iso}}^{1/4} \sim \delta_T^{1/8}$ and, via an adapted Magnanini–Poggesi oscillation bound, a Hausdorff annulus width $\eta \sim \varepsilon^\theta + r_* \sim \delta_T^{1/8}$. This is *not* (9): $\delta_T^{1/8} \gg \delta_T^{1/4}$. The remaining gap is therefore *one power*, $\delta_T^{1/8} \rightarrow \delta_T^{1/4}$, i.e. lowering the exterior fjord scale from $\delta_{\text{iso}}^{1/4}$ to $\delta_{\text{iso}}^{1/2}$ — a Magnanini–Poggesi oscillation bound under the Serrin density, one power sharper than what the flux mechanism currently yields. That single estimate is all that stands between this and a fully selection-free sharp theorem.

Proposition 13 (Unconditional selection-free bound). *For every bounded $\Omega \subset \mathbb{R}^2$ with $\delta_T(\Omega) \leq \delta_*$,*

$$\mathcal{A}_F(\Omega)^2 \leq C \delta_T(\Omega)^{2/3},$$

with no selection, no regularity, and no inclusion hypothesis.

Proof. Pick the good level at removed volume λ (Lemma of §Step 2): $D_I(\hat{t}) \leq 16\pi\delta_T/\lambda$ and $|\Omega \setminus E| = \lambda$, $E = \{u > \hat{t}\} \subset \Omega$. The inset's isoperimetric deficit obeys $\delta_{\text{iso}}(E) \lesssim D_I(\hat{t}) \lesssim \delta_T/\lambda$, so by the sharp quantitative isoperimetric inequality (FMP), $\mathcal{A}_F(E) \lesssim \sqrt{\delta_{\text{iso}}(E)} \lesssim \sqrt{\delta_T/\lambda}$. Comparing Ω to the volume- π ball concentric with E 's optimal ball, $\mathcal{A}_F(\Omega) \leq \mathcal{A}_F(E) + 2\lambda/\pi \lesssim \sqrt{\delta_T/\lambda} + \lambda$. Optimising $\lambda = \delta_T^{1/3}$ gives $\mathcal{A}_F(\Omega) \lesssim \delta_T^{1/3}$. $\square$

Remark 14 (Why this beats the inclusion bound). The Hausdorff-annulus route gives only $\mathcal{A}_F(\Omega) \leq 2\rho_e\eta \sim \delta_T^{1/8}$, i.e. $\mathcal{A}_F^2 \lesssim \delta_T^{1/4}$ — *worse* than Proposition 13 and conditional on the inclusion. The point is that the *inset's* perimeter is controlled by δ_T (unlike $\partial\Omega$'s), so FMP applies to it directly. The graph routes exist to remove the FMP square-root loss ($\mathcal{A}_F \sim \sqrt{\delta_{\text{iso}}}$) and reach the sharp $\mathcal{A}_F^2 \lesssim \delta_T$.

8 An overhang-free alternative: the Steiner flow

If the $\delta_T^{1/4}$ annulus proves stubborn, a structurally different route avoids inclusions altogether. Brock's continuous Steiner symmetrization Ω_t^e ($t \in [0, 1]$, direction e) is monotone for torsion and *overhang-free by construction* (after symmetrizing in e , every line $\parallel e$ meets the set in one segment).

Proposition 15 (Monotone torsion, dissipation, deficit splitting). *With $u = u_\Omega \geq 0$ extended by 0 and u_t^e its continuous Steiner rearrangement, $t \mapsto T(\Omega_t^e)$ is non-decreasing and*

$$T(\Omega_t^e) \geq 2 \int u_t^e - \int |\nabla u_t^e|^2 \geq T(\Omega), \quad \mathcal{D}^e(\Omega) := \int |\nabla u|^2 - \int |\nabla u^e|^2 \geq 0.$$

For a director schedule $\{e_k\}$ whose iterated symmetrizations converge in L^1 to B_1 (e.g. equidistributed; $[4, 5]$), $\delta_T(\Omega) = \sum_k \mathcal{D}^{e_k}(\Omega_{k-1})$.

Proof. $u_t^e \in H_0^1(\Omega_t^e)$ (super-level sets symmetrize into Ω_t^e), so u_t^e is admissible in $T(\Omega_t^e) = \max\{2 \int g - \int |\nabla g|^2\}$; the Pólya–Szegő content of Brock's theorem $\int |\nabla u_t^e|^2 \leq \int |\nabla u|^2$ with $\int u_t^e = \int u$ gives the displayed chain. Telescoping the energy drops along the schedule and $T = \int |\nabla u|^2$ on torsion functions yields the splitting. $\square$

The sharp selection-free, overhang-free theorem then follows from a quantitative Pólya–Szegő $\mathcal{D}^e(\Omega) \geq c \|u - u^e\|_{L^1}^2$ plus an isotropy averaging — the torsion analogues of the Dolbeault–Esteban–Figalli–Frank–Loss estimates [6], neither of which uses selection or regularity. This is recorded as the backup; the optimal-graph route of §5 is the primary one because it is elementary modulo the single inclusion estimate.

9 Summary

1. **Target (yours).** It suffices to realize Ω by a radial graph G with controlled volume and deficit; closeness to Ω is then automatic, and the radial-graph theorem closes (1) (Theorem 2). The graph theorem needs amplitude $\leq$ a *fixed constant* — automatic. **No $\sqrt{\delta_T}$ -width annulus is required**; that was an error in my previous note.
2. **Cut, do not fill.** The radial hull fails (overhangs give defect $\gg \delta_T$). The optimal radial graph, which cuts thin caps, has $\text{gd}(\Omega) \lesssim \eta(\delta_{\text{iso}} + \eta)$ (Theorem 4) — *with no no-overhang hypothesis*. Overhangs are charged to δ_{iso} .
3. **What the surgery costs.** Of the four hypotheses of Theorem 2, the graph realizes amplitude, volume, and closeness *for free*; only the deficit (iii) is constrained, the surgical repair shifting δ_T by $\lesssim \eta^2\delta_{\text{iso}}$. This is $O(\delta_T)$ iff $\eta \lesssim \delta_T^{1/4}$ (Corollary 6).

4. **Reduction (Route 2).** Hence, selection-free / $C^{1,\gamma}$ -free / overhang-free, $\mathcal{A}_F^2 \lesssim \delta_T$ once $\eta \lesssim \delta_T^{1/4}$ (Theorem 7).
5. **Route 1 (no surgery).** The folding identity $\int_0^\infty F(t) dt = \int_\Omega (\partial_r u)_+$ (Lemma 10) shows: the inset is a graph iff $\partial_r u \leq 0$ in the boundary layer (radial monotonicity). The interior holds for free (Lemma 11); the obstruction is the *sign of $\partial_r u$ in the layer* (11). The same $M = \int (\partial_r u)_+$ governs both routes: Route 1 is $M = 0$, Route 2 pays $\propto M$. Pure Route 1 is of the same strength as the boundary regularity selection supplies.
6. **Status of the inputs.** Route 2's Hausdorff annulus width: with selection it is BNST; without, flux/density gives $\delta_T^{1/8}$, one power short of $\delta_T^{1/4}$ (gap = fjord scale $\delta_{\text{iso}}^{1/4} \rightarrow \delta_{\text{iso}}^{1/2}$). Route 1's input is the radial-monotonicity sign condition.
7. **Unconditional.** With no selection and no inclusion, $\mathcal{A}_F^2 \lesssim \delta_T^{2/3}$, by FMP applied to the inset (Proposition 13).
8. **Backup.** The continuous Steiner flow is overhang-free and splits the deficit (Proposition 15), reducing the sharp theorem to a DEFFL-type quantitative Pólya–Szegő.

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